

Assembly Line — Machine Profile Sheet

Machine Profile Sheet · Quick Reference

Doc ID	PROF-ASM-003
Version	1.0
Effective Date	2026-03-15
Owner	Manufacturing Engineering
Applies To	Assembly Line — Plant B (Chennai), Plant C (Ahmedabad); units Assembly_Line_1–20

1. Capacity

Metric	Value
Fleet size (this machine type)	20 units
Average ideal cycle time	1.40 min/part
Theoretical rated output	43 parts/hour — 344 parts/shift (480 min, no losses)

2. Rated Speed / Mechanical Specification

Parameter	Specification
Conveyor speed	0.2 – 0.6 m/s, variable
Station count	6 fitment stations per line
Torque tools	Calibrated pneumatic torque wrenches, ±3% accuracy
Fixture actuation	Shared hydraulic lift/tilt power unit

3. Typical OEE Target

Industry world-class benchmark for discrete manufacturing is 85% OEE (90% availability × 95% performance × 99% quality). Measured performance for this machine type across the current dataset is **88.2% OEE** (availability 95.3%, performance 94.4%, quality 98.0%). Plant target for this machine type is set at ≥85% OEE, with any sustained reading below 80% triggering a reliability review under Plant_Maintenance_Policy.

4. Critical Components

- Shared hydraulic lift/tilt power unit — largest downtime contributor, see DT-01.
- Conveyor drive motor and chain — tracking and jam-related stoppages (DT-05).
- Pneumatic torque tools — calibration drift risk (DT-06), directly tied to fastening quality.
- Station sensors / limit switches — highest-frequency minor stoppage source (DT-04).

5. Common Failure Modes (ranked by share of logged downtime)

Failure Mode	Code	Share of Downtime	Avg Duration
Hydraulic System Failure	DT-01	67%	145 min avg

Failure Mode	Code	Share of Downtime	Avg Duration
Raw Material / Feed Shortage	DT-02	17%	49 min avg
Tooling / Die Changeover Delay	DT-03	11%	37 min avg
Spindle / Main Bearing Failure	DT-08	5%	63 min avg